

SESSION 22

Mathematical Optimization

Convex vs Non-Convex Functions · Global & Local Extrema · Gradient Descent Algorithms

Introduction to Mathematical Optimization

Convexity & Convex Optimization

Global vs Local Extrema

Saddle Points & Plateau Regions

Gradient Descent Algorithm

Learning Rate Optimization

Gradient Descent Variants (GD, SGD, MBGD)

Modern Optimizers (Adam, RMSprop)

Session Notes — Complete Reference

Playlist: Maths for Machine Learning · Instructor: Nitish Singh (CampusX)



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01 • Introduction to Mathematical Optimization

Optimization is the process of finding the inputs that minimize or maximize an objective function (e.g., minimizing a loss function in machine learning).

02 · Convexity & Convex Optimization

A function is **convex** if a line segment drawn between any two points on the graph lies above or on the graph. Convex functions are easier to optimize because any local minimum is guaranteed to be the global minimum.

03 · Global vs Local Extrema

- **Global Minimum:** The absolute lowest value of the function over its entire domain.
- **Local Minimum:** The lowest value of the function within a specific local neighborhood.

04 · Saddle Points & Plateau Regions

A **saddle point** is a point where the function's gradient is zero, but it is not a local extremum (it curves up in one direction and down in another). Saddle points are major obstacles when training deep neural networks.

05 • Gradient Descent Algorithm

Gradient Descent is an iterative optimization algorithm that updates parameters in the direction of the steepest descent (opposite to the gradient ∇f):

GRADIENT DESCENT STEP

$\theta_{\text{(new)}} = \theta_{\text{(old)}} - \eta \cdot \nabla f(\theta_{\text{(old)}})$ where η is the learning rate.

06 · Learning Rate Optimization

- **Too small learning rate:** Very slow convergence.
- **Too large learning rate:** Overshoots the minimum and may diverge.

07 · Gradient Descent Variants (GD, SGD, MBGD)

VARIANT	BATCH SIZE	UPDATES PER EPOCH	CONVERGENCE BEHAVIOR
Batch GD	Full dataset	1	Smooth and stable, but slow
Stochastic GD (SGD)	1 sample	N	Noisy, helps escape local minima
Mini-Batch GD	Batch size \$\$\$	N/B	Best balance of speed and stability

08 · Modern Optimizers (Adam, RMSprop)

Modern deep learning optimizers adapt the learning rate for each parameter using running averages of the gradients (momentum) and squared gradients (RMSprop).

09 · Quick Reference Card



Optimization Cheat Sheet

- **Gradient Direction:** $\nabla f(\theta)$ points in direction of steepest ascent.
- **Convex Function:** Local minimum = Global minimum.
- **SGD:** Updates parameters using a single random sample at a time.